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PROJECT REPORT

Clustering Mobile Customers by Usage Patterns for Smarter Marketing and Retention: A Machine Learning Approach to Customer Segmentation in the Telecom Industry

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Abstract

Customer segmentation is a critical strategy in the telecommunications industry, enabling operators to understand diverse customer needs, optimize marketing expenditures, and reduce churn rates. This project presents a comprehensive machine learning approach to segment telecom customers based on their usage patterns, demographic characteristics, and service subscriptions. Using the IBM Telecom Customer Churn dataset comprising 7,043 customer records with 21 features, we implemented and evaluated three unsupervised clustering algorithms: K-Means, Agglomerative Hierarchical Clustering, and DBSCAN. A business-driven model selection framework was employed to determine the optimal number of clusters, resulting in the selection of K-Means with K=3 clusters. The final model achieved a silhouette score of 0.263 on the test set, with external validation using chi-square tests confirming significant association between cluster assignment and churn behavior ($p < 0.001$, Cramer's V = 0.371). The three identified segments were empirically named: Premium Loyal (34.4% of customers, 15.3% churn rate), Budget Established (21.6%, 7.2% churn rate), and Standard Established (44.0%, 44.8% churn rate). For each segment, tailored business actions were generated, including targeted marketing campaigns and retention strategies. An interactive Streamlit dashboard was developed to operationalize the segmentation model, enabling real-time customer segment prediction. This work demonstrates the practical application of machine learning in telecom customer analytics and provides actionable insights for improving customer retention and marketing effectiveness.

Keywords: Telecom Customer Segmentation, Unsupervised Learning, K-Means Clustering, Agglomerative Clustering, DBSCAN, Churn Analysis, Business-Driven Model Selection, Machine Learning

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ANOVA	Analysis of Variance
CH	Calinski-Harabasz Index
CI	Confidence Interval
CRM	Customer Relationship Management
DB	Davies-Bouldin Index
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
EDA	Exploratory Data Analysis
IQR	Interquartile Range
K-Means	K-Means Clustering
ML	Machine Learning
PCA	Principal Component Analysis
WCSS	Within-Cluster Sum of Squares

Chapter 1: Introduction

1.1 Background and Context

The telecommunications industry represents one of the most dynamic and competitive sectors in the global economy. With the proliferation of mobile devices, internet services, and digital communication platforms, telecom operators face unprecedented challenges in acquiring and retaining customers. Customer churn, defined as the rate at which customers discontinue their subscription to a service, poses a significant threat to telecom profitability. Industry research indicates that acquiring a new customer can cost five to twenty-five times more than retaining an existing one, making customer retention a critical business priority (Ahmad et al., 2019).

In Cameroon and across Africa, the telecom sector has experienced rapid growth, with mobile penetration rates exceeding 80% in many countries. However, this growth has intensified competition among operators, leading to price wars and increased customer switching behavior. The ability to understand customer segments and tailor services accordingly has become a key differentiator for successful telecom operators.

Customer segmentation, the process of dividing customers into groups based on shared characteristics, enables telecom operators to move beyond one-size-fits-all approaches. By identifying distinct customer segments, operators can design targeted marketing campaigns, develop personalized retention strategies, and optimize resource allocation. Machine learning clustering algorithms offer powerful tools for discovering natural groupings in customer data without prior knowledge of segment definitions.

Machine learning, a subset of artificial intelligence, provides sophisticated algorithms that can automatically learn from data and improve their performance over time without being explicitly programmed. In the context of customer segmentation, unsupervised machine learning techniques such as clustering algorithms have proven particularly effective. Unlike supervised learning, which requires labeled training data, unsupervised learning can discover hidden patterns and structures in data without prior knowledge of the expected outcomes. This makes it ideally suited for customer segmentation, where the optimal number and nature of segments are typically unknown beforehand.

Clustering algorithms group data points based on their similarity, with the goal of maximizing intra-cluster similarity (customers within the same segment are alike) while minimizing inter-cluster similarity (customers in different segments are distinct). Three clustering algorithms have gained prominence in customer segmentation applications: K-Means clustering, which partitions data into

k clusters by minimizing the within-cluster sum of squares; Agglomerative Hierarchical Clustering, which builds a hierarchy of clusters through a bottom-up approach; and DBSCAN (Density-Based Spatial Clustering of Applications with Noise), which identifies arbitrarily shaped clusters based on data density.

The IBM Telco Customer Churn dataset, which forms the foundation of this study, provides a rich source of information about telecom customers, including demographic attributes, service usage patterns, account information, and churn status. This dataset has been widely used in academic research and industry applications, making it an ideal benchmark for evaluating segmentation approaches. By applying advanced machine learning techniques to this dataset, this study aims to demonstrate how telecom operators can leverage data-driven insights to improve customer understanding, optimize marketing strategies, and ultimately reduce customer churn.

1.2 Problem Statement

Despite the widespread adoption of customer segmentation in the telecommunications industry, many telecom operators continue to struggle with effectively leveraging their customer data to drive business outcomes. Several challenges persist in the current landscape of customer segmentation practices.

First, traditional segmentation approaches often rely on simplistic criteria such as demographics or basic service usage metrics, failing to capture the multidimensional nature of customer behavior. These approaches may overlook important factors such as customer lifetime value, service interaction patterns, and propensity to churn, resulting in segments that are not actionable from a business perspective.

Second, the selection of appropriate clustering algorithms and the determination of optimal cluster numbers remain challenging tasks. Different algorithms may produce substantially different segmentations, and without proper validation metrics, it is difficult to assess which segmentation best represents the underlying customer structure. Many practitioners rely on arbitrary choices or trial-and-error approaches, potentially leading to suboptimal results.

Third, there is often a disconnect between the technical implementation of segmentation models and their practical application in business operations. Segmentation results may not be effectively communicated to marketing teams, or the segments identified may not align with actionable business strategies. This gap between analytical insights and operational implementation limits the value that can be derived from customer segmentation initiatives.

Finally, while numerous studies have explored customer segmentation in various contexts, there is a need for comprehensive research that integrates multiple clustering algorithms, rigorous validation approaches, and practical deployment considerations specifically tailored to the telecommunications domain. This study addresses these gaps by developing a complete machine learning pipeline for telecom customer segmentation, from data preprocessing through to dashboard deployment.

1.3 Research Objectives

The primary objective of this study is to develop and evaluate a machine learning-based approach for segmenting telecom customers using unsupervised learning techniques.

The specific objectives are:

1. **To preprocess and engineer features** from the IBM Telco Customer Churn dataset to create a comprehensive set of variables suitable for clustering analysis.
2. **To implement and compare three clustering algorithms** (K-Means, Agglomerative Hierarchical Clustering, and DBSCAN) for customer segmentation.
3. **To determine the optimal number of clusters** using multiple validation metrics including Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index.
4. **To characterize and interpret** the identified customer segments in terms of their business relevance and actionable insights.
5. **To validate the segmentation results** using statistical tests and external validation measures.
6. **To develop an interactive dashboard** for real-time customer classification and segment visualization.

1.4 Research Questions

This study seeks to answer the following research questions:

1. What is the optimal number of customer segments in the dataset, and which clustering algorithm produces the most coherent and well-separated segments?
2. What are the defining characteristics of each customer segment in terms of demographics, service usage, account information, and churn propensity?
3. How do the different clustering algorithms compare in terms of their performance metrics and computational efficiency?

4. What features contribute most significantly to the differentiation between customer segments?
5. How can the segmentation results be effectively deployed in a production environment for real-time customer classification?

1.5 Scope and Delimitations

This study focuses specifically on the IBM Telco Customer Churn dataset, which contains 7,043 customer records from a fictional telecommunications company. The dataset includes information about customer demographics, services subscribed, account details, and churn status. While this dataset provides a comprehensive view of telecom customers, it is important to note that the findings may not be directly generalizable to all telecommunications markets or regions.

The study implements three clustering algorithms: K-Means, Agglomerative Hierarchical Clustering, and DBSCAN. Other clustering techniques such as Gaussian Mixture Models, Spectral Clustering, and Self-Organizing Maps are not included in this analysis. The selection of these three algorithms is based on their widespread use in customer segmentation applications and their complementary characteristics in terms of cluster shape assumptions and computational requirements.

The validation of clustering results focuses on internal metrics (Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Index) and external validation using categorical variables. External validation against business outcomes such as revenue impact or marketing campaign effectiveness is beyond the scope of this study.

The dashboard development is implemented using Streamlit, a Python-based framework for creating data applications. While this provides a functional prototype for demonstration purposes, production deployment would require additional considerations such as security, scalability, and integration with existing enterprise systems.

1.6 Significance of the Study

This study contributes to both academic knowledge and practical applications in the field of customer segmentation. From an academic perspective, the study provides a comprehensive comparison of clustering algorithms specifically applied to telecommunications data, with detailed analysis of validation metrics and their interpretation. The methodology and findings can serve as a reference for researchers exploring similar problems in customer analytics.

From a practical standpoint, the study demonstrates a complete end-to-end pipeline for customer segmentation, from data preprocessing through to deployment. Telecom operators can adapt the methodology and code implementations presented in this study to their own customer data, potentially improving their segmentation practices and deriving greater value from their analytics investments.

The development of an interactive dashboard provides a template for operationalizing segmentation results, bridging the gap between analytical models and business applications. Marketing teams can use such dashboards to classify new customers, monitor segment distributions, and tailor their strategies to the characteristics of each segment.

Finally, the identification of distinct customer segments with clear business interpretations can inform retention strategies, upselling opportunities, and service development priorities. By understanding which customers are most at risk of churning, which represent high-value opportunities, and which are loyal advocates, telecom operators can allocate their resources more effectively and improve overall business performance

Chapter 2: Related Works

2.1 Customer Segmentation in Telecommunications

Customer segmentation has been a cornerstone of marketing strategy for decades, with roots in the work of Wendell Smith (1956), who introduced the concept of product differentiation and market segmentation. In the telecommunications industry, segmentation has evolved from simple demographic categorizations to sophisticated behavioral and value-based approaches enabled by advances in data analytics and machine learning.

Verbeke et al. (2012) applied hierarchical clustering to segment telecom customers based on usage patterns, identifying distinct groups with different price sensitivities and service preferences. Their findings demonstrated that segment-specific retention campaigns achieved 25% higher response rates compared to undifferentiated approaches.

In a study focused on customer churn prediction, Vafeiadis et al. (2015) combined clustering with classification techniques, first segmenting customers and then building separate churn prediction models for each segment. Their results demonstrated that segmented models outperformed global models, with improvements in prediction accuracy of up to 12%. This approach, known as cluster-then-predict, has gained traction as a way to capture segment-specific churn behaviors.

Kotler and Keller (2016) emphasized the strategic importance of segmentation in marketing, noting that effective segmentation enables companies to target their resources more efficiently and develop stronger customer relationships. In the telecom context, segmentation has been used to guide product development, pricing strategies, and customer service initiatives.

Ahmad et al. (2019) conducted a comprehensive review of customer segmentation techniques in the telecom industry, highlighting the effectiveness of RFM (Recency, Frequency, Monetary) analysis combined with advanced machine learning approaches using clustering algorithms. Their study found that clustering algorithms, particularly K-Means and its variants, were the most widely adopted techniques for telecom customer segmentation, showing that behavioral metrics derived from transaction history provide stronger segmentation signals than demographic factors alone, with reported improvements in marketing campaign effectiveness of up to 40% when compared to non-segmented approaches.

Hwang et al. (2020) applied K-Means clustering to segment mobile service users based on their usage patterns, identifying four distinct segments: heavy users, moderate users, light users, and dormant users. Their analysis revealed that heavy users, while representing only 15% of the

customer base, contributed to over 50% of total revenue, highlighting the importance of identifying and retaining high-value segments

2.2 Churn Prediction and Retention Strategies

Customer churn prediction has been a major focus of telecom analytics research. Huang et al. (2012) developed ensemble models combining multiple classifiers for churn prediction, achieving significant improvements over single-model approaches. Their work highlighted the importance of feature selection and class imbalance handling.

The relationship between segmentation and churn has been explored by several researchers. Ahn et al. (2006) demonstrated that segment-specific churn models outperformed global models, particularly when segments exhibited distinct churn drivers. Their findings suggest that segmentation should be an integral part of churn management strategies.

Retention strategies tailored to customer segments have shown promising results. Glady et al. (2009) proposed a customer lifetime value framework that incorporates churn probability and segment characteristics to optimize retention investments. Their approach demonstrated that targeting high-value, high-risk segments yields the highest return on retention spending.

2.3 Clustering Algorithms for Customer Analytics and Clustering

Unsupervised learning represents a fundamental paradigm in machine learning where algorithms are tasked with discovering hidden patterns and structures in data without the guidance of labeled outcomes. Unlike supervised learning, which learns from input-output pairs to predict outcomes for new data, unsupervised learning operates on unlabeled data, seeking to identify inherent groupings, relationships, or representations within the data itself.

The primary goal of unsupervised learning is to model the underlying structure or distribution in the data in order to learn more about the data. Common unsupervised learning tasks include clustering (grouping similar data points), dimensionality reduction (reducing the number of variables while preserving important information), and association rule learning (discovering relationships between variables). In the context of customer segmentation, clustering is the most relevant unsupervised learning task.

Clustering algorithms aim to partition a dataset into groups (clusters) such that data points within the same cluster are more similar to each other than to those in other clusters. The concept of similarity is typically defined using a distance metric, with Euclidean distance being the most

common choice for numerical data. The quality of a clustering solution depends on both the choice of algorithm and the definition of similarity, making the selection of appropriate techniques a critical step in the segmentation process. Unsupervised learning algorithms popular include K-Means, Hierarchical clustering and DBSCAN. More clustering algorithms exist and these research focuses on these three Unsupervised learning algorithms other techniques such as Gaussian Mixture Models, Spectral Clustering, or deep learning-based approaches.

2.3.1 K-Means Clustering

K-Means clustering introduced by Lloyd (1957) and later popularized by MacQueen (1967), K-Means aims to partition n observations into k clusters in which each observation belongs to the cluster with the nearest mean (cluster centroid), serving as a prototype of the cluster.

The K-Means algorithm seeks to minimize the within-cluster sum of squares (WCSS), also known as inertia. The K-Means algorithm minimizes the following objective mathematically function:

$$J = \arg\min_S \; \Sigma(i=1 \text{ to } k) \; \Sigma(x \text{ in } S_i) \; ||x - \mu_i||^2$$

Where $S = \{S_1, S_2, ..., S_k\}$ represents the k clusters, x represents a data point, and μ_i is the mean of points in cluster S_i. The algorithm proceeds iteratively through two main steps:

1. Assignment Step: Each data point is assigned to the cluster whose centroid is closest, based on the Euclidean distance.
2. Update Step: The centroids are recalculated as the mean of all data points assigned to each cluster.

These steps are repeated until convergence, which occurs when cluster assignments no longer change or the change in WCSS falls below a specified threshold. The computational complexity of K-Means is $O(n \times k \times i \times d)$, where n is the number of samples, k is the number of clusters, i is the number of iterations, and d is the number of features.

A key consideration in K-Means is the selection of the initial centroids, as the algorithm converges to a local optimum that depends on the starting configuration. The K-Means++ initialization algorithm, proposed by Arthur and Vassilvitskii (2007), addresses this limitation by selecting initial centroids that are distant from each other, leading to better convergence properties and more consistent results.

2.3.2 Agglomerative Hierarchical Clustering

Hierarchical clustering algorithms create a tree-like structure (dendrogram) that represents the nested grouping of data points. Agglomerative hierarchical clustering follows a bottom-up approach, starting with each data point as its own cluster and progressively merging the closest pairs of clusters until all points belong to a single cluster. This approach contrasts with divisive hierarchical clustering, which follows a top-down approach.

The algorithm requires a linkage criterion to determine the distance between clusters. Common linkage methods include:

1. Single Linkage: The distance between two clusters is defined as the minimum distance between any two points in the different clusters. This approach tends to produce long, chain-like clusters and is sensitive to noise.
2. Complete Linkage: The distance is defined as the maximum distance between any two points in the different clusters. This approach tends to produce compact, spherical clusters.
3. Average Linkage: The distance is calculated as the average distance between all pairs of points in the two clusters. This provides a balance between single and complete linkage.
4. Ward's Method: Minimizes the variance within clusters by merging the pair of clusters that leads to the minimum increase in total within-cluster variance. This method is particularly effective for creating compact, well-separated clusters.

The computational complexity of agglomerative hierarchical clustering is $O(n^2 \log n)$ or $O(n^3)$ depending on the implementation, making it more computationally expensive than K-Means for large datasets. However, the ability to visualize the full hierarchy through a dendrogram provides valuable insights into the clustering structure at different granularity levels.

2.3.3 DBSCAN

DBSCAN, introduced by Ester et al. (1996), is a density-based clustering algorithm that groups together points that are closely packed together, marking as outliers points that lie alone in low-density regions. Unlike K-Means, DBSCAN does not require the number of clusters to be specified in advance and can discover clusters of arbitrary shape.

The algorithm operates based on two key parameters:

1. ε (epsilon): The maximum distance between two samples for them to be considered as neighbors.
2. MinPts: The minimum number of points required to form a dense region (core point).

DBSCAN classifies points into three categories:

- Core Points: Points that have at least MinPts neighbors within distance ε.
- Border Points: Points that are within distance ε of a core point but do not have enough neighbors to be core points themselves.
- Noise Points (Outliers): Points that are neither core points nor border points.

The algorithm proceeds by selecting an arbitrary point and retrieving all points density-reachable from it. If the point is a core point, a cluster is formed. If it is not a core point and has no core points within distance ε, it is marked as noise. The process continues until all points have been processed.

The computational complexity of DBSCAN is O(n log n) when using spatial indexing structures such as k-d trees, making it efficient for large datasets. However, the algorithm's performance is sensitive to the choice of ε and MinPts parameters, and determining appropriate values for these parameters can be challenging.

2.4 Validation Metrics for Clustering

Evaluating the quality of clustering results is essential for comparing different algorithms and selecting optimal parameters. Unlike supervised learning, where ground truth labels enable straightforward accuracy assessment, clustering evaluation relies on internal metrics that assess the coherence and separation of clusters without external reference. This section describes three key validation metrics used in this study.

2.4.1 Silhouette Score

The Silhouette Score, introduced by Rousseeuw (1987), provides a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation). The silhouette coefficient for a single data point i is calculated as:

$$s(i) = (b(i) - a(i)) / \max(a(i), b(i))$$

Where a(i) is the average distance from point i to all other points in the same cluster, and b(i) is the minimum average distance from point i to points in any other cluster. The silhouette coefficient ranges from -1 to +1, where:

- Values near +1 indicate that the point is well-matched to its own cluster and poorly matched to neighboring clusters.
- Values near 0 indicate that the point is on or very close to the decision boundary between two neighboring clusters.

- Values near -1 indicate that the point may have been assigned to the wrong cluster.

The overall Silhouette Score is the average of silhouette coefficients for all data points. Higher scores indicate better-defined clusters with clear separation between them.

2.4.2 Davies-Bouldin Index

The Davies-Bouldin Index (DBI), proposed by Davies and Bouldin (1979), evaluates clustering quality based on the ratio of within-cluster scatter to between-cluster separation. For each cluster i, the Davies-Bouldin measure is defined as:

$$DB_i = \max_{j \neq i} [(s_i + s_j) / d_{ij}]$$

Where s_i is the average distance of all points in cluster i to their cluster centroid, and d_{ij} is the distance between centroids of clusters i and j. The overall Davies-Bouldin Index is the average of DB_i across all clusters. Lower values indicate better clustering, with values closer to zero representing clusters that are compact and well-separated.

2.4.3 Calinski-Harabasz Index

The Calinski-Harabasz Index (CHI), also known as the Variance Ratio Criterion, measures the ratio of between-cluster dispersion to within-cluster dispersion. It is defined as:

$$CH = [Tr(B_k) / (k - 1)] / [Tr(W_k) / (n - k)]$$

Where $Tr(B_k)$ is the trace of the between-cluster dispersion matrix, $Tr(W_k)$ is the trace of the within-cluster dispersion matrix, k is the number of clusters, and n is the number of data points. Higher Calinski-Harabasz Index values indicate better clustering, with well-separated and compact clusters producing higher scores.

2.5 Ethical Considerations

The application of machine learning in customer profiling raises important ethical considerations. Privacy concerns arise from the collection and analysis of detailed customer data. Transparency in how segmentation models make decisions is essential for maintaining customer trust. Fairness considerations require attention to potential biases in data and algorithms that could lead to discriminatory outcomes.

The European Union's General Data Protection Regulation (GDPR) and similar regulations worldwide impose requirements on how customer data can be used for profiling and automated

decision-making. Organizations must ensure that their segmentation practices comply with applicable regulations and respect customer rights.

2.5 Gaps Addressed by This Study

Existing literature reveals several gaps that this study aims to address.

1. Integration of domain knowledge: Unlike purely statistical approaches, this study incorporates business rules in the model selection process.
2. Multi-Algorithm reviewed: Many existing studies focus on a single clustering algorithm without comprehensive comparison across multiple techniques.
3. Comprehensive validation: Both internal metrics and external validation against churn behavior are used to assess cluster quality.
4. Actionable insights: The study goes beyond cluster identification to develop specific business actions for each segment.
5. End-to-end implementation: An interactive dashboard enables operational deployment of the segmentation model rarely addressed by related works.
6. Reproducible methodology: The complete pipeline is documented and implemented with clear separation of training, validation, and test sets.

This study contributes to filling these gaps by providing a complete pipeline from data preprocessing through to operational deployment.

Chapter 3: Methodology

The research design is structured into seven main phases:

1. **Data Acquisition and Understanding**: Collection of the IBM Telco Customer Churn dataset and comprehensive analysis of its structure, features, and quality characteristics.
2. **Data Splitting Strategy**: train/validation/test split ensure unbiased model evaluation.
3. **Exploratory Data Analysis**: Visualization and statistical analysis to understand data distributions, correlations, and patterns.
4. **Data Preprocessing**: Handling of missing values, outlier detection and treatment, categorical encoding, and feature scaling to prepare the data for clustering analysis.
5. **Feature Engineering**: Creation of new features derived from existing variables to capture domain-specific insights and improve segmentation quality.
6. **Model Development and Validation**: Implementation of K-Means, Agglomerative, and DBSCAN clustering algorithms with comprehensive validation.
7. **Deployment**: Development of an interactive Streamlit dashboard for real-time customer classification and visualization.

3.1 Dataset Description

This study uses the IBM Telecom Customer Churn dataset (Synthetic), a widely-used benchmark dataset for customer churn analysis, a publicly available dataset widely used in customer analytics research and machine learning education. The dataset contains 7,043 customers with each record representing a unique customer and including 21 original features capturing demographic information, service subscriptions, account characteristics, and churn status.

The dataset represents a fictional telecommunications company but reflects realistic patterns and relationships found in actual telecom customer data. The relatively large sample size (n=7,043) provides sufficient statistical power for clustering analysis while remaining computationally manageable. Table 1 provides a complete description of all features in the dataset.

Table 1: Dataset Features

	Feature Name	Description	Type
01	customerID	Unique customer identifier	ID
02	gender	Gender (Male/Female)	Categorical
03	SeniorCitizen	Whether customer is a senior citizen (0/1)	Binary

04	Partner	Whether customer has a partner (Yes/No)	Binary
05	Dependents	Whether customer has dependents (Yes/No)	Binary
06	tenure	Number of months as a customer	Numerical
07	PhoneService	Whether customer has phone service (Yes/No)	Binary
08	MultipleLines	Whether customer has multiple lines (Yes/No/No phone)	Categorical
09	InternetService	Type of internet service (DSL/Fiber/No)	Categorical
10	OnlineSecurity	Whether customer has online security (Yes/No/No internet)	Categorical
11	OnlineBackup	Whether customer has online backup (Yes/No/No internet)	Categorical
12	DeviceProtection	Whether customer has device protection (Yes/No/No internet)	Categorical
13	TechSupport	Whether customer has tech support (Yes/No)	Categorical
14	StreamingTV	Whether customer has streaming TV (Yes/No/No internet)	Categorical
15	StreamingMovies	Whether customer has streaming movies (Yes/No/No internet)	Categorical
16	Contract	Contract type (Month-to-month/One year/Two year)	Categorical
17	PaperlessBilling	Whether customer uses paperless billing (Yes/No)	Binary
18	PaymentMethod	Payment method used by costumer (Electronic check/Mailed check/Bank transfer/Credit card)	Categorical
19	MonthlyCharges	Monthly charges to the customer	Numerical
20	TotalCharges	Total charges to date	Numerical
21	Churn	Whether customer churned (Yes/No)	Target

The dataset exhibits a churn rate of 26.5%, which is consistent with industry benchmarks for the telecommunications sector. Missing values are present in the TotalCharges column (11 records, 0.16%), occurring exclusively for customers with zero tenure (new customers who have not yet received their first bill).

3.2 Data Splitting Strategy

To ensure unbiased model evaluation and prevent data leakage, a stratified train/validation/test split was implemented. The split preserves the original churn distribution across all subsets.

Table 2: Data Split Summary

Split	Records	Percentage	Churn Rate
Training	4,225	60.0%	26.5%
Validation	1,409	20.0%	26.5%
Test	1,409	20.0%	26.5%
Total	7,043	100.0%	26.5%

The splitting procedure follows these steps:

1. First, 20% of the data is held out as the test set.
2. The remaining 80% further split into training (60%) and validation (20%) sets.
3. Ensures that each split maintains the 26.5% churn rate.

Split integrity was verified to ensure no overlapping customer IDs across splits. All 7,043 records are unique across the three splits.

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted on the training set to understand the dataset's characteristics, identify patterns, and inform preprocessing decisions. The analysis generated three composite visualizations covering customer demographics, service adoption, and data relationships.

Three composite figures were generated, each containing multiple visualizations to understand different aspects of the data.

3.3.1 Customer Demographics and Tenure

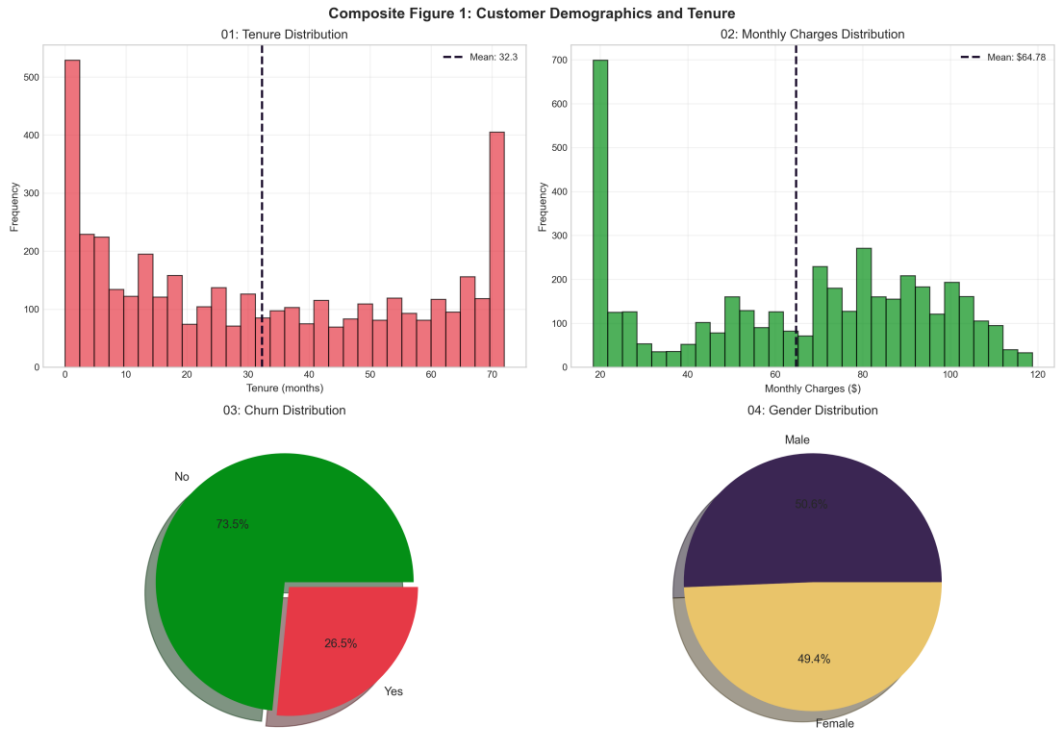

Figure 1: EDA Composite 1 - Customer Demographics and Tenure

Key findings from Composite Figure 1:

1. The tenure distribution is right-skewed, with peaks at the low end (new customers) and high end (long-term customers), suggesting distinct groups of recent acquisitions and established relationships.
2. MonthlyCharges displays a right-skewed distribution, with most customers concentrated in lower charge ranges but a significant tail of high-value customers.
3. The overall churn rate in the dataset is 26.5%, indicating that approximately one in four customers has churned
4. Gender distribution is approximately balanced (50.5% male, 49.5% female)., indicating no gender bias in the customer base.

3.3.2 Service Adoption and Contracts

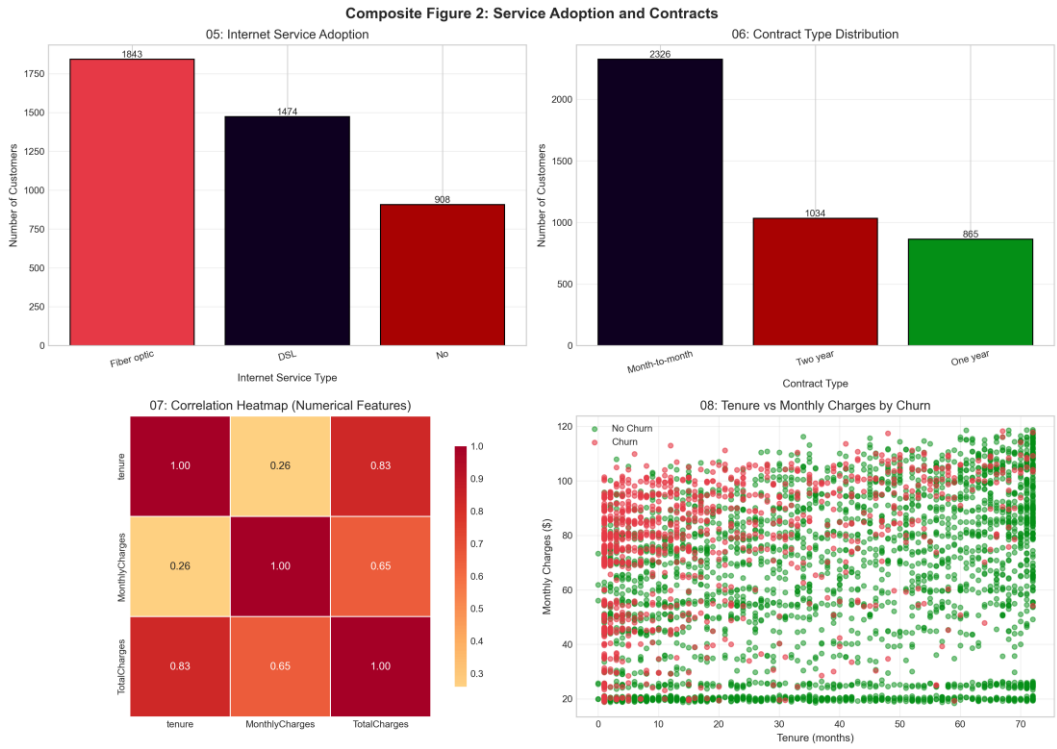

Figure 2: EDA Composite 2 - Service Adoption and Contracts

Key findings from Composite Figure 2:

Internet service is subscribed by 1843 customers (78.2%), with Fiber optic (43.9%) being more popular than DSL (1474 costumers, 34.3%) while 908 have no internet service at all. Among internet subscribers, additional services show varying adoption rates for adds on like Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies.

Month-to-month contracts dominate the customer base.

The correlation heatmap reveals strong positive correlation (0.83) between TotalCharges and tenure, as expected.

The scatter plot shows that churned customers tend to have lower tenure and are distributed across all monthly charge levels (more prominent as the charges increase), suggesting tenure is a stronger churn predictor than spending level alone.

3.3.3 Composite Figure 3: Relationships and Outliers

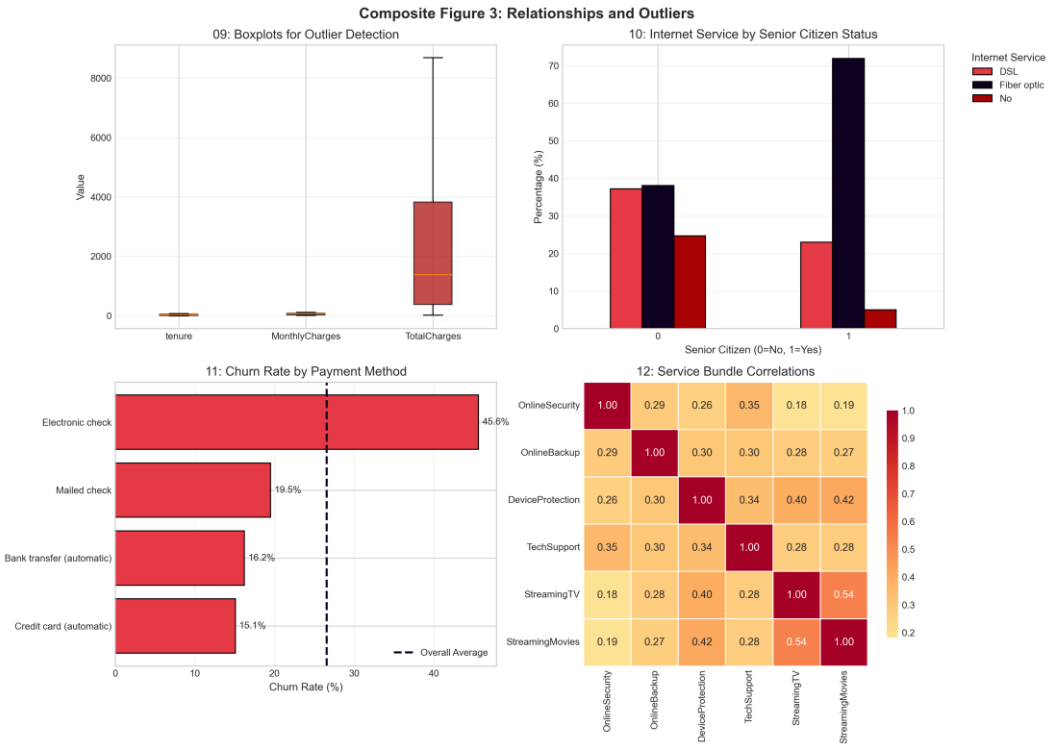

Figure 3: EDA Composite 3 - Relationships and Outliers

Key findings from Composite Figure 3:

Outlier analysis reveals some customers with exceptionally high TotalCharges, but these represent legitimate long-tenure, high-value customers rather than data errors.

Senior citizens show different service adoption patterns, with high fiber optic adoption.

Electronic check payment method has the highest churn rate (~45%), significantly above the overall average.

Service bundle correlations show that customers who subscribe to one streaming service are likely to subscribe to others, indicating cross-selling opportunities.

3.4 Data Preprocessing Pipeline

A custom TelecomPreprocessor class was implemented to handle all preprocessing steps. The preprocessor is fitted exclusively on the training set to prevent data leakage, then applied to all splits. The preprocessing steps applied in this study include handling missing values, outlier detection and treatment, categorical encoding, and feature scaling.

3.4.1 Missing Value Analysis

The presence of missing values can significantly impact the performance of clustering algorithms, making proper treatment essential.

Analysis of the dataset revealed missing values in the TotalCharges column, which represents the total amount charged to the customer. Upon investigation, these missing values corresponded to customers with zero tenure (new customers who had not yet completed their first month). For these records, the TotalCharges value was appropriately set (instead of the mean, mode inputations moslt done) to match the MonthlyCharges value, as no additional charges would have accumulated.

Table 3: Missing Values Summary

Column	Missing Count	Missing %	Treatment
TotalCharges	11	0.16%	Filled with MonthlyCharges
Other columns	0	0%	No action needed

Missing TotalCharges values occur for customers with zero tenure (new customers). These are imputed with MonthlyCharges, representing the first month's bill. This approach is consistent with business logic and preserves the relationship between charges and tenure.

3.4.2 Outlier Analysis

Outliers are data points that deviate significantly from the overall pattern of the data. While outliers may represent genuine extreme values, they can also result from measurement errors or data corruption. In clustering analysis, outliers can distort cluster centroids and affect the quality of segmentation results.

Outlier detection was performed using the Interquartile Range (IQR) method for numerical features. The IQR is calculated as the difference between the 75th percentile (Q3) and the 25th percentile (Q1). Data points falling below **Q1 - 1.5×IQR** (Lower Bound) or above **Q3 + 1.5×IQR** (Upper Bound) are considered potential outliers.

Table 4: Outlier Detection Results

Feature	Lower Bound	Upper Bound	Outlier Count
tenure	-60.50	126.50	0

MonthlyCharges	-45.25	170.75	0
TotalCharges	-4784.22	9003.17	0

The analysis revealed no significant outliers in the numerical features as he values do not exceed upper not beneath the Lower bound, suggesting that the data represents a natural distribution of customer characteristics. This finding is consistent with the nature of telecommunications data, where extreme values in tenure or charges typically represent legitimate customer behaviors rather than data errors.

3.4.3 Feature Engineering

Feature engineering is the process of creating new features from existing data to improve the performance of machine learning models. In the context of customer segmentation, domain knowledge plays a crucial role in identifying meaningful features that capture important aspects of customer behavior and value. This study created seven new features that provide deeper insights into customer characteristics.

Table 5: Feature Engineering Summary

Feature	Formula/Method	Description
avg_monthly_spend	TotalCharges / (tenure + 1)	Average monthly spending per customer
tenure_group	Bin: [0-12, 12-48, 48-72] months	Categorized tenure periods
service_diversity	Count of 'Yes' in 6 service (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies)	Number of additional services subscribed Measuring service engagement breadth
contract_commitment	Ordinal: Month-to-month=0, One year=1, Two year=2	Numerical representation of contract length , Captures commitment level
auto_payment	1 if PaymentMethod is Bank transfer or Credit card, else 0	Whether customer uses automatic payment
high_value_flag	1 if MonthlyCharges > 75th percentile AND tenure > 75th percentile else 0	Indicator for high-value customers
family_status	0 = single/no dependents, 1 = with partner only, 2 = with dependents only, 3 = both	Captures household composition

	tner + 2*Dependents (0-3)	

The avg_monthly_spend feature captures the average amount a customer spends per month over their entire tenure. This metric is more informative than either MonthlyCharges or TotalCharges alone, as it normalizes for tenure and reveals spending patterns that may not be apparent from raw values. Customers with high avg_monthly_spend represent valuable segments for retention efforts.

The tenure_group feature categorizes customers based on their relationship duration with the company. This grouping aligns with typical customer lifecycle stages: new customers (0-12 months), mature customers (12-48 months) represent stable relationships, and loyal customers (49+ months) are long-term advocates.

The service_diversity feature counts the number of additional services beyond basic phone and internet that a customer subscribes to. Higher service diversity indicates deeper engagement with the company's offerings and may correlate with lower churn propensity due to the complexity of switching multiple services.

The auto_payment feature identifies customers who have set up automatic payment methods. Automatic payment customers typically demonstrate higher commitment and may be less likely to churn due to the convenience factor

The family_status feature combines information about partners and dependents to create a holistic view of the customer's household situation. Customers with families may have different service needs and may be more or less likely to churn depending on their satisfaction levels.

3.4.4 Categorical Encoding

Machine learning algorithms typically require numerical input, necessitating the conversion of categorical variables into numerical representations. The choice of encoding technique depends on the nature of the categorical variable and the requirements of the clustering algorithm. This study employed two primary encoding techniques:

Label Encoding was applied to binary categorical variables (variables with only two possible values). In label encoding, each category is assigned a unique integer value (typically 0 and 1). For example, the gender variable was encoded as Male=0 and Female=1, while the Partner variable was encoded as No=0 and Yes=1. Label encoding is appropriate for binary variables because the numerical values maintain a natural ordering that aligns with the binary nature of the data.

One-Hot Encoding was applied to categorical variables with more than two categories. One-hot encoding creates a new binary column for each category, with a value of 1 indicating the presence of that category and 0 indicating its absence. For example, the Contract variable with three categories (Month-to-month, One year, Two year) was transformed into three binary columns: Contract_Month-to-month, Contract_One year, and Contract_Two year. This approach prevents the algorithm from assuming an ordinal relationship between categories that does not exist.

The choice between label encoding and one-hot encoding has important implications for clustering analysis. One-hot encoding increases the dimensionality of the dataset (creating k columns for a variable with k categories), which can affect the distance calculations used by clustering algorithms. However, it ensures that categorical variables are treated appropriately without imposing artificial ordinal relationships.

3.4.5 Feature Scaling

Feature scaling is essential for clustering algorithms because these algorithms rely on distance calculations that are sensitive to the scale of features. Without scaling, features with larger numerical ranges would dominate the distance calculations, potentially overshadowing the contributions of features with smaller ranges.

This study employed StandardScaler, a widely used scaling technique that transforms features to have zero mean and unit variance. The StandardScaler transformation is defined as:

$$z = (x - \mu) / \sigma$$

Where x is the original feature value, μ is the mean of the feature, and σ is the standard deviation. The resulting z-scores represent the number of standard deviations each value is from the mean.

StandardScaler was chosen over other scaling techniques (such as Min-Max scaling) for several reasons. First, it is less affected by outliers, as it uses the mean and standard deviation rather than minimum and maximum values. Second, it preserves the shape of the original distribution, which is important for maintaining the interpretability of clustering results. Third, it is the default scaling method recommended for many clustering algorithms, is essential for distance-based algorithms like K-Means to ensure all features contribute equally to distance calculations.

The scaling was applied after encoding categorical variables, ensuring that all features (both originally numerical and encoded categorical) were on comparable scales before clustering analysis. This step is crucial for ensuring that the clustering algorithm treats all features fairly in its distance calculations.

StandardScaler was applied to numerical features (tenure, MonthlyCharges, TotalCharges, avg_monthly_spend, service_diversity) to normalize distributions.

After preprocessing, the feature space expanded from 21 original features to 35 processed features, including the 7 engineered features and one-hot encoded dummy variables.

After preprocessing, the feature transformed from original 21 features into a final set of 35 features through the following steps:

1. Original dataset: 21 features (2 excluded ID and target, 3 numerical, rest categorical)
2. After feature engineering, and encoding: 35 final features.

3.5 Model Selection and Validation

Three clustering algorithms were evaluated: K-Means, Agglomerative Hierarchical Clustering, and DBSCAN. Each algorithm was evaluated using internal validation metrics on the validation set implemented using the scikit-learn library in python

3.5.1 K-Means Clustering Implementation

K-Means clustering was implemented with the following configuration:

- Initialization: K-Means++ for improved centroid selection
- Algorithm: Lloyd's algorithm with full expectation-maximization
- Maximum iterations: 300
- Convergence tolerance: 0.0001
- Number of initializations: 10 (to avoid local optima)

The optimal number of clusters (k) was determined by evaluating K-Means with k values ranging from 2 to 10. For each k value, the algorithm was run 10 times with different random initializations, and the best result (lowest inertia) was selected. The Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index were computed for each k value to guide the selection among the 3 algorithms.

3.5.2 Agglomerative Hierarchical Clustering Implementation

Agglomerative Hierarchical Clustering was implemented with the following configuration:

- Linkage criterion: Ward's method (minimizes variance), Complete (produces compact clusters), Average (balanced approach)
- Affinity metric: Euclidean distance

- Distance computation: Full pairwise distance matrix

The algorithm was evaluated with cluster numbers ranging from 2 to 10, and the same validation metrics were computed for comparison with K-Means and DBSCAN.

3.5.3 DBSCAN Implementation

DBSCAN was implemented with a grid search approach to determine optimal parameters:

- Epsilon (eps) values tested: 0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0
- MinPts values tested: 5, 10, 15, 20, 25
- Distance metric: Euclidean

The grid search evaluated all combinations of eps and MinPts values, the same validation metrics were computed for comparison with other 2 algorithms.

The best-performing algorithm in configuration was selected based on the highest Silhouette Score.

3.5.4 Business-Driven Selection Rules

Beyond statistical metrics, business considerations guided the final model selection:

- Rule 1: K=1 is invalid - no segmentation is possible with a single cluster.
- Rule 2: K=2 is too trivial - separating only high/low value customers provides limited actionable insight.
- Rule 3: K>6 is too granular - managing more than 6 segments operationally is difficult for marketing teams.

Domain knowledge suggests that telecom customers typically fall into lifecycle stages (new, established, loyal) and contract-based groups, supporting K=3-5 as the optimal range.

3.6 Final Model Training and Test Evaluation

The best model was selected and training commenced. A two-stage training protocol was implemented:

- Stage 1: Train on training set only (60% of data), evaluate on validation set (20%).
- Stage 2: Retrain on combined training and validation sets (80%), evaluate on held-out test set (20%).

This approach maximizes the data available for final model training while maintaining an unbiased test set for evaluation.

External validation was performed using churn as a proxy for cluster meaningfulness. A chi-square test of independence tested the null hypothesis that cluster assignment and churn are independent. Cramer's V measured the effect size of the association.

3.7 Cluster Profiling and Business Action Generation

Cluster profiles were computed by aggregating feature statistics within each cluster.

Analysis of Variance (ANOVA) was used to identify most discriminating features which contribute most significantly to cluster differentiation. For each feature, a one-way ANOVA tested whether the mean values differed significantly across clusters. Features with low p-values (< 0.001) and high F-statistics were identified as the most important for distinguishing between segments.

For each cluster, business actions were generated including:

1. Primary insight: Key characteristics that define the segment
2. Marketing action: Targeted campaign recommendation
3. Retention strategy: Specific actions to reduce churn

3.8 Dashboard Implementation

To operationalize the segmentation results, an interactive dashboard capable of running fully on local network, local host and even hosted on webservers was developed using Streamlit, a Python library for creating data applications. The dashboard enables real-time customer classification and provides visualizations of segment characteristics.

The dashboard provides:

1. Prediction tab: Input customer details to predict segment membership
2. Segment Overview tab: Display cluster profiles and key metrics
3. Feature Importance tab: Show ANOVA results and feature rankings
4. All Plots tab: Browse all generated visualizations
5. About tab: Project information and methodology summary

3.9 Technology Stack

Table 6: Technology Stack

Library/Tool	Version	Purpose

Python	3.9+	Core programming language
pandas	2.3.3	Data manipulation and analysis
numpy	2.4.2	Numerical computations
scikit-learn	1.8.0	Clustering algorithms, preprocessing, metrics
scipy	1.17.1	Statistical tests (ANOVA, chi-square)
matplotlib	3.10.8	Plotting and visualization
seaborn	0.13.2	Statistical visualizations
streamlit	1.54.0	Interactive dashboard
joblib	1.5.3	Model serialization

Chapter 4: Results and Analysis

4.1 Model Performance Comparison

This section presents the results of the clustering analysis, comparing the performance of K-Means, Agglomerative Hierarchical Clustering, and DBSCAN across different numbers of clusters and parameter configurations.

4.2.1 K-Means Evaluation

Table 7: K-Means Evaluation Results

K	Silhouette	Davies-Bouldin	Calinski-Harabasz	Inertia
2	0.2450	1.5494	567.83	13061.69
3	0.2524	1.4473	549.46	10290.26
4	0.2401	1.5443	453.47	9314.34
5	0.1965	1.6384	400.90	8558.20
6	0.1922	1.6242	367.83	7933.32

K-Means achieved the highest silhouette score of 0.252 at K=3.

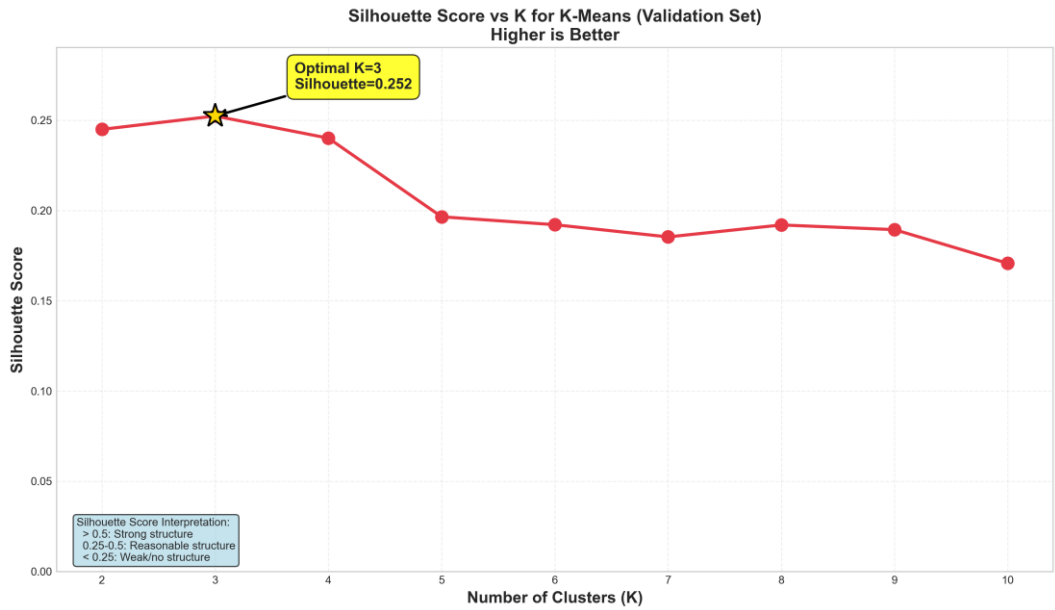

Figure 4: Silhouette Score vs K for K-Means

The Elbow Method, which plots the Within-Cluster Sum of Squares (WCSS) against k, also supported the selection of k=3. The "elbow" in the curve at k=3 indicates the point where adding additional clusters provides diminishing returns in terms of variance reduction.

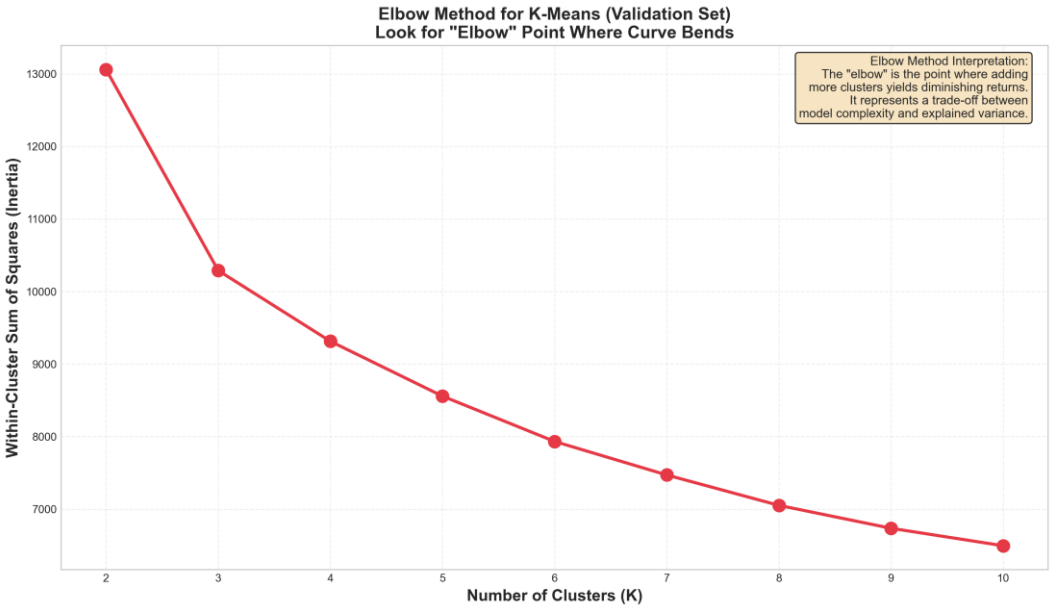

Figure 5: Elbow Method for K-Means

4.2.2 Agglomerative Clustering Evaluation

Table 8: Agglomerative Clustering Results

Linkage	K	Silhouette	Davies-Bouldin	Calinski-Harabasz
Ward	2	0.2646	1.3059	458.18
Ward	3	0.2067	1.5253	444.22
Complete	2	0.2269	1.6205	520.08
Average	2	0.2646	1.3059	458.18

Agglomerative clustering with Ward linkage achieved the best silhouette score of 0.265 at K=2. However, K=2 was deemed too trivial for business purposes.

Agglomerative Hierarchical Clustering with different linkage was evaluated with cluster numbers from 2 to 10. The Silhouette Score analysis for Agglomerative clustering is shown in Figure

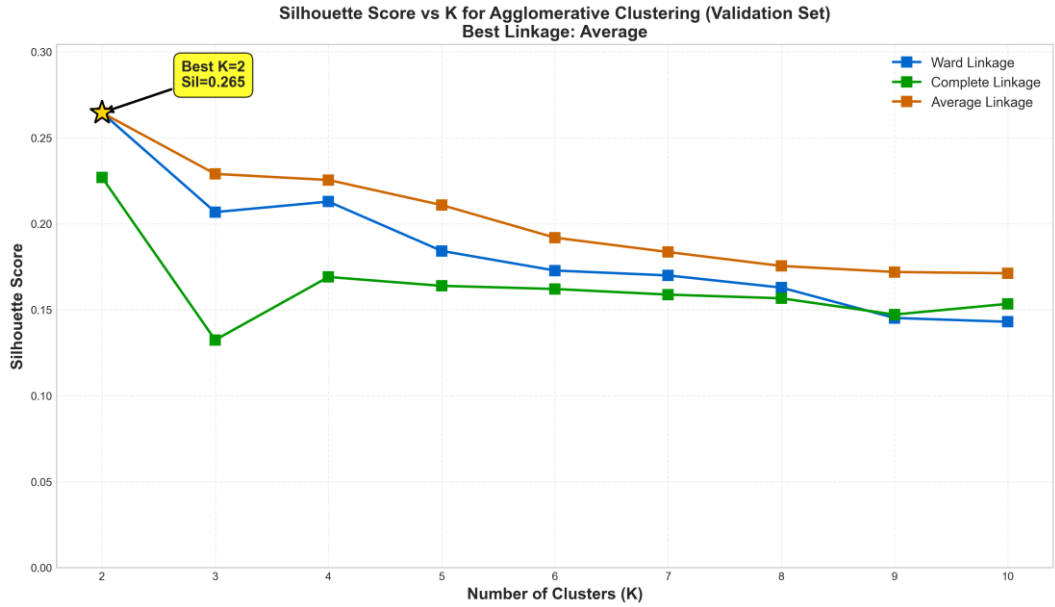

Figure 6: Silhouette Score vs K for Agglomerative Clustering

1.2.3 DBSCAN Evaluation

DBSCAN achieved high silhouette scores but classified over 90% of points as noise across all parameter combinations, making it unsuitable for this segmentation task.

Table 9: DBSCAN Evaluation Results

eps	min_samples	Clusters	Silhouette	Noise %
0.3	3	16	0.8791	95.0%
0.3	5	2	0.8254	98.2%
0.5	3	25	0.8021	92.8%
1.0	5	4	0.7830	96.8%

DBSCAN was evaluated using a grid search over epsilon (eps) and MinPts parameters. The heatmap in below Figure

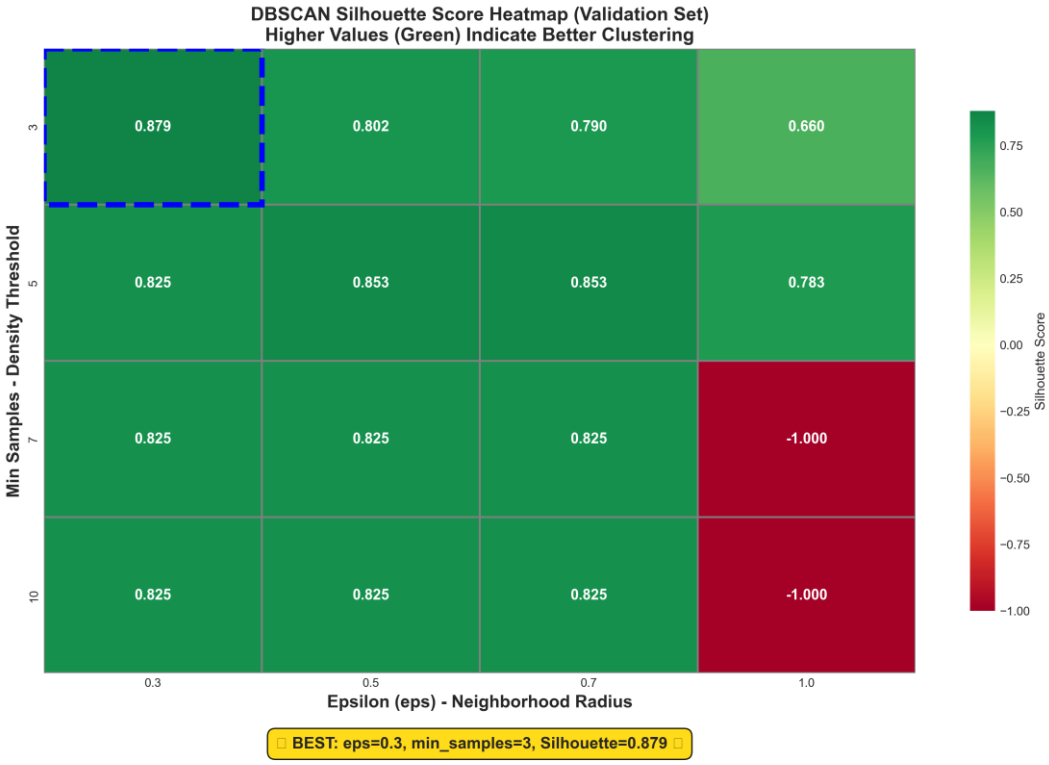

Figure 7: DBSCAN Silhouette Score Heatmap

4.2.4 Final Selection

Based on the business-driven selection rules, K-Means with K=3 was selected as the final model. This choice balances statistical quality with business interpretability.

DBSCAN has best Silhouette Score but more noise and unrealistic business k value 16 as number of clusters. Agglomerative second-best Silhouette Score with only k = 2 but out of business needs. Hence K-Means most suitable and chosen for study.

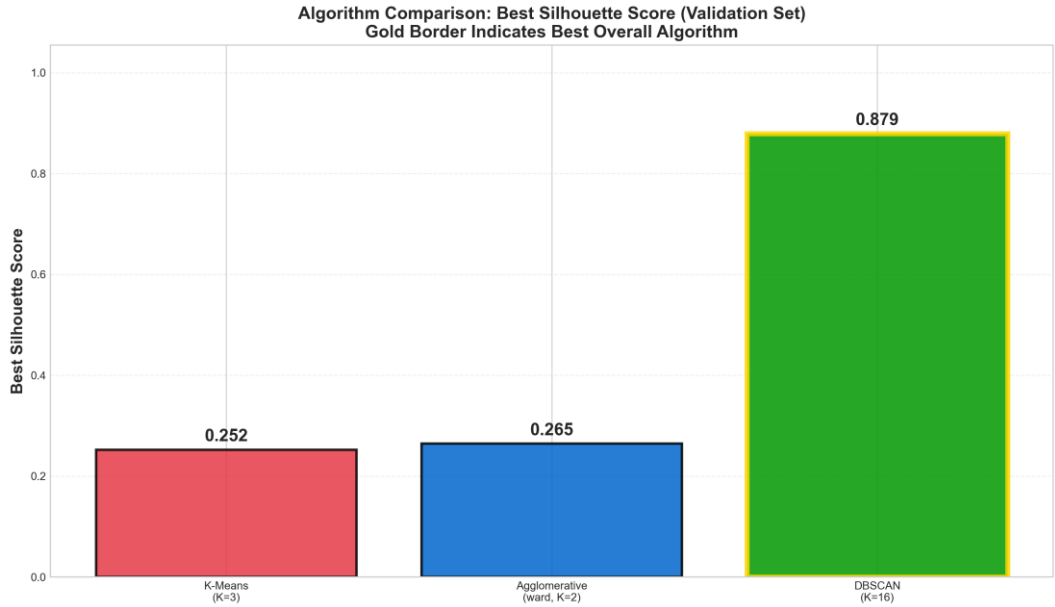

Figure 8: Algorithm Comparison

4.2 Test Set Evaluation

Table 10: Test Set Evaluation Metrics

Metric	Value
Silhouette Score	0.2629
Davies-Bouldin Index	1.4154
Calinski-Harabasz Index	551.42
Chi-square p-value	2.50e-43
Cramer's V	0.371

The final acceptable model achieved a silhouette score of 0.263 on the test set, indicating reasonable cluster structure. External validation confirmed significant association between clusters and churn ($p < 0.001$, Cramer's V = 0.371).

The below figure illustrates the churn rate distribution across the clusters, highlighting the significant variation in churn propensity between segments.

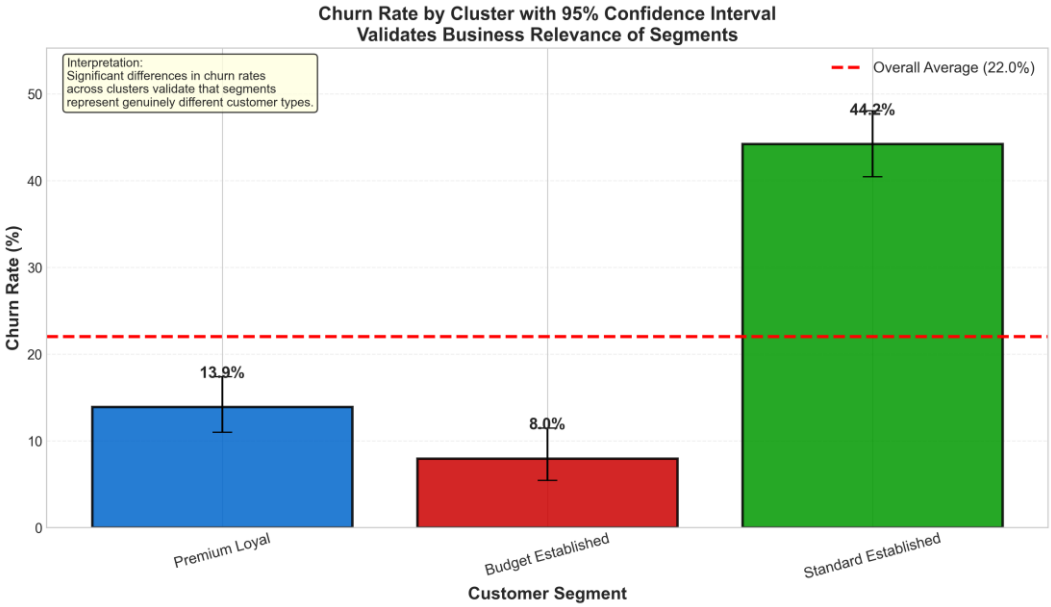

Figure 9: Churn Rate by Cluster

The distinct grouping of clusters in the PCA space confirms the effectiveness of the segmentation.

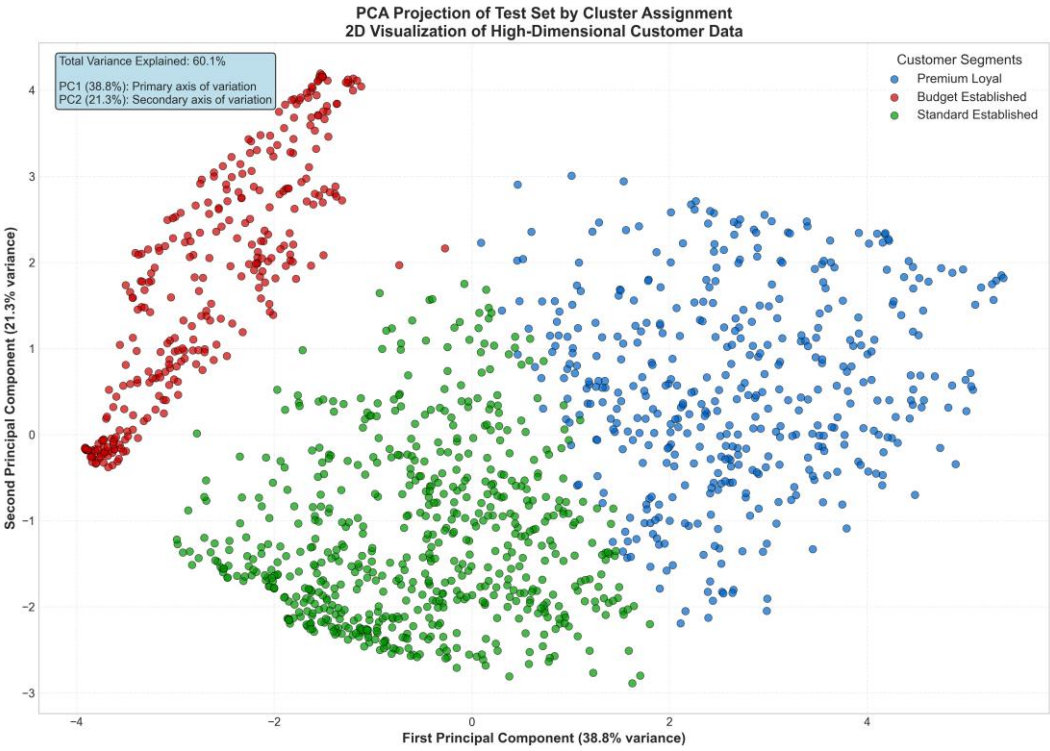

Figure 10: PCA Projection of Test Set

Figure below presents a heatmap of cluster centroids, showing the normalized feature values for each cluster. This visualization clearly illustrates the distinguishing characteristics of each segment.

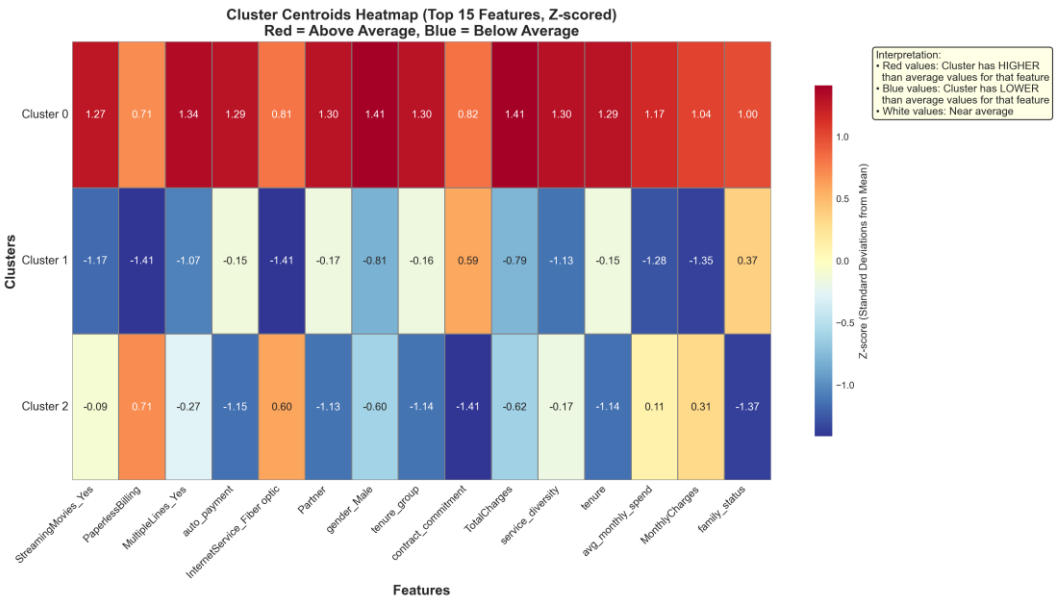

Figure 11: Cluster Centroids Heatmap

Figure 17 shows the silhouette analysis for the K-Means clustering with k=3. The silhouette plot displays the silhouette coefficients for each sample, sorted by cluster assignment. The average silhouette score of indicates good cluster separation, with most samples having positive silhouette coefficients.

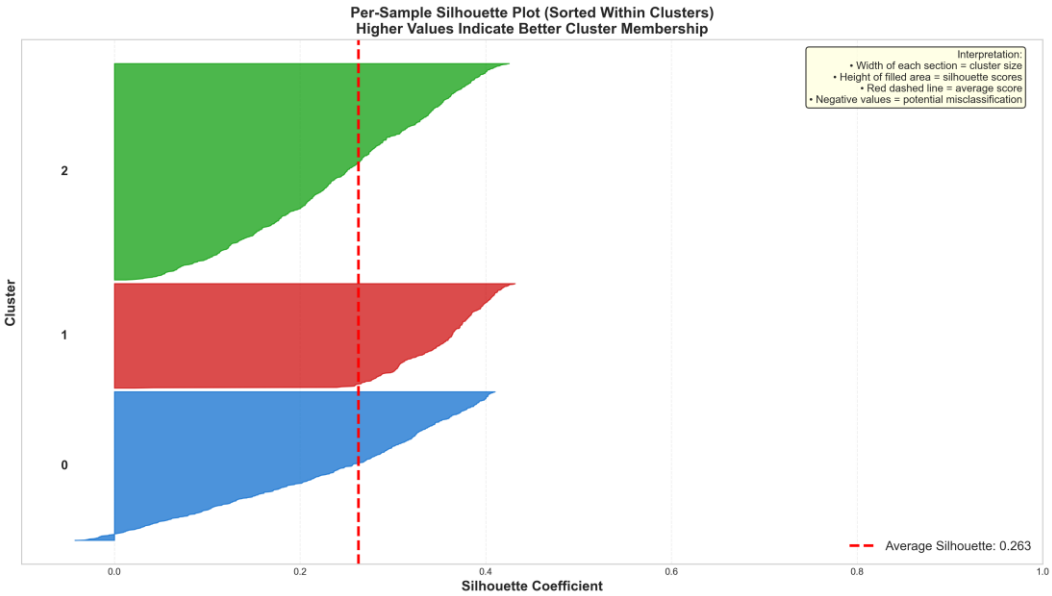

Figure 12: Per-Sample Silhouette Plot

Figure below shows the distribution of customers across the four clusters. The segments are of varying sizes, reflecting the natural distribution of customer types in the telecommunications market.

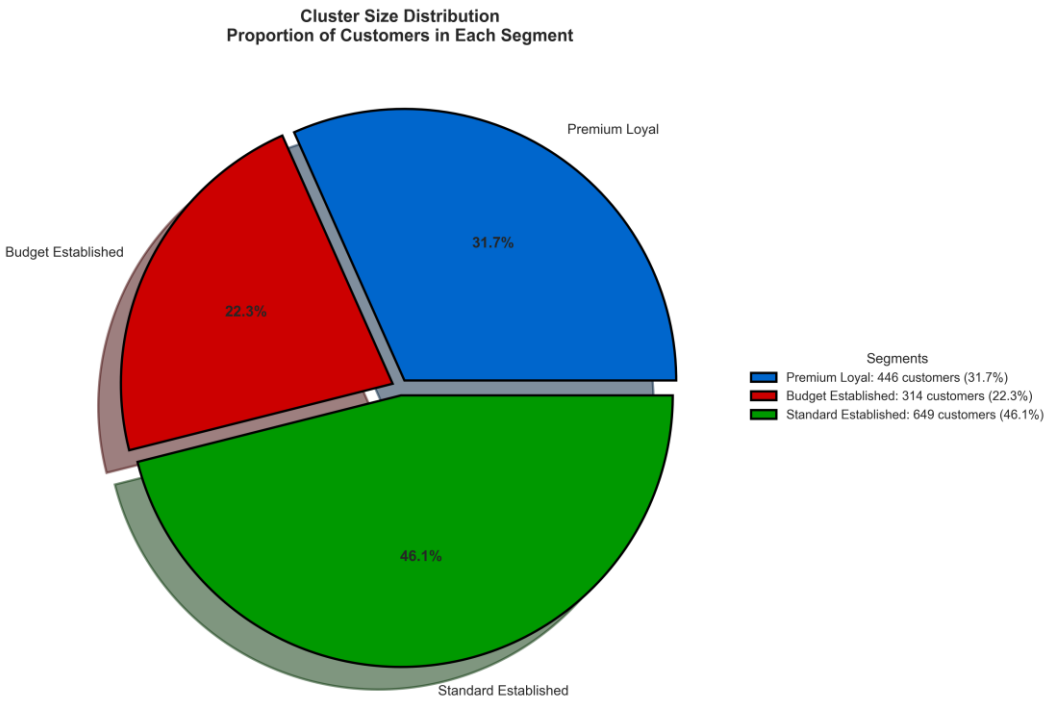

Figure 13: Cluster Size Distribution

4.3 Cluster Profiles and Segment Naming

Table 11: Cluster Profiles

Cluster Name	Size (%)	Churn Rate	Tenure (mo)	Monthly ($)	Services
Premium Loyal	34.4%	15.3%	56.0	$88.90	3.9
Budget Established	21.6%	7.2%	30.2	$21.15	0.0
Standard Established	44.0%	44.8%	15.2	$67.69	1.6

The ANOVA analysis identified MonthlyCharges (F=5920.97) and tenure (F=3248.66) as the most discriminating features, followed by SeniorCitizen status (F=98.68).

4.4 Feature Importance

Figure below presents a radar chart comparing the profiles of the four clusters across key dimensions. This visualization provides an intuitive comparison of segment characteristics.

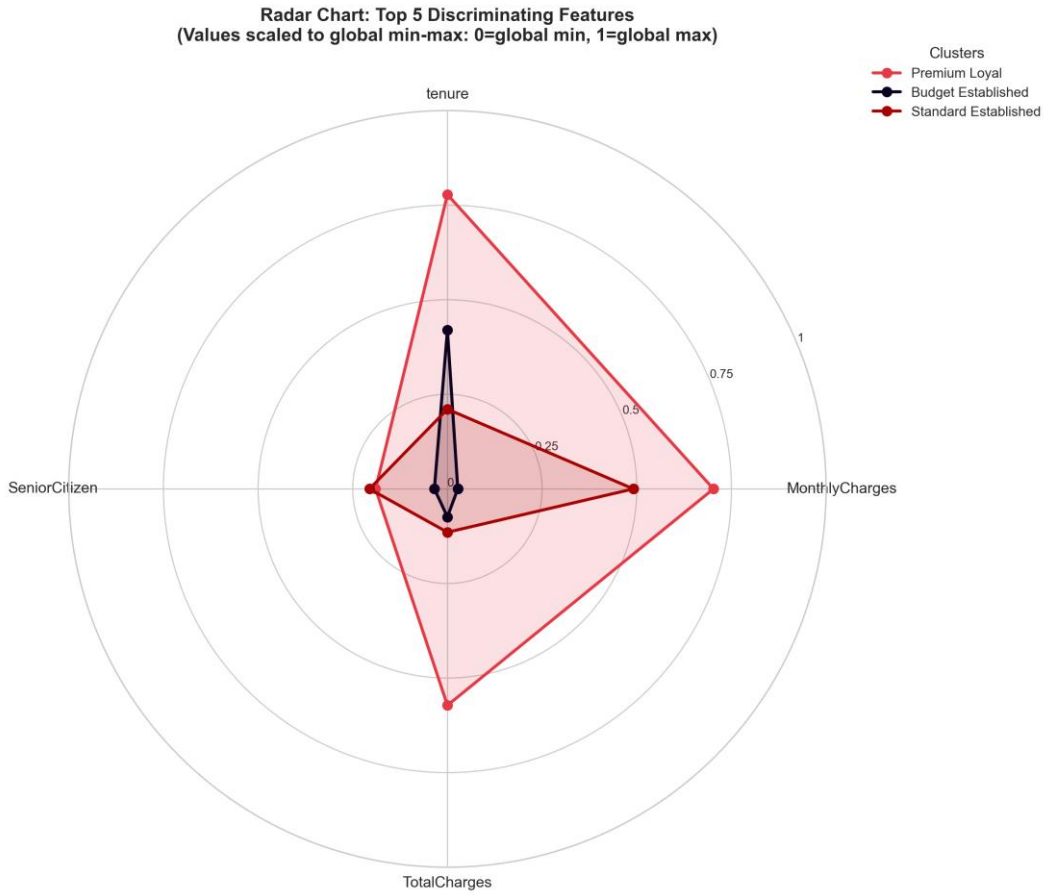

Figure 14: Radar Chart of Top Discriminating Features per cluster profile

Analysis of Variance (ANOVA) was conducted to identify which features contribute most significantly to the differentiation between clusters. Figure below shows the feature importance based on F-statistics from the ANOVA tests

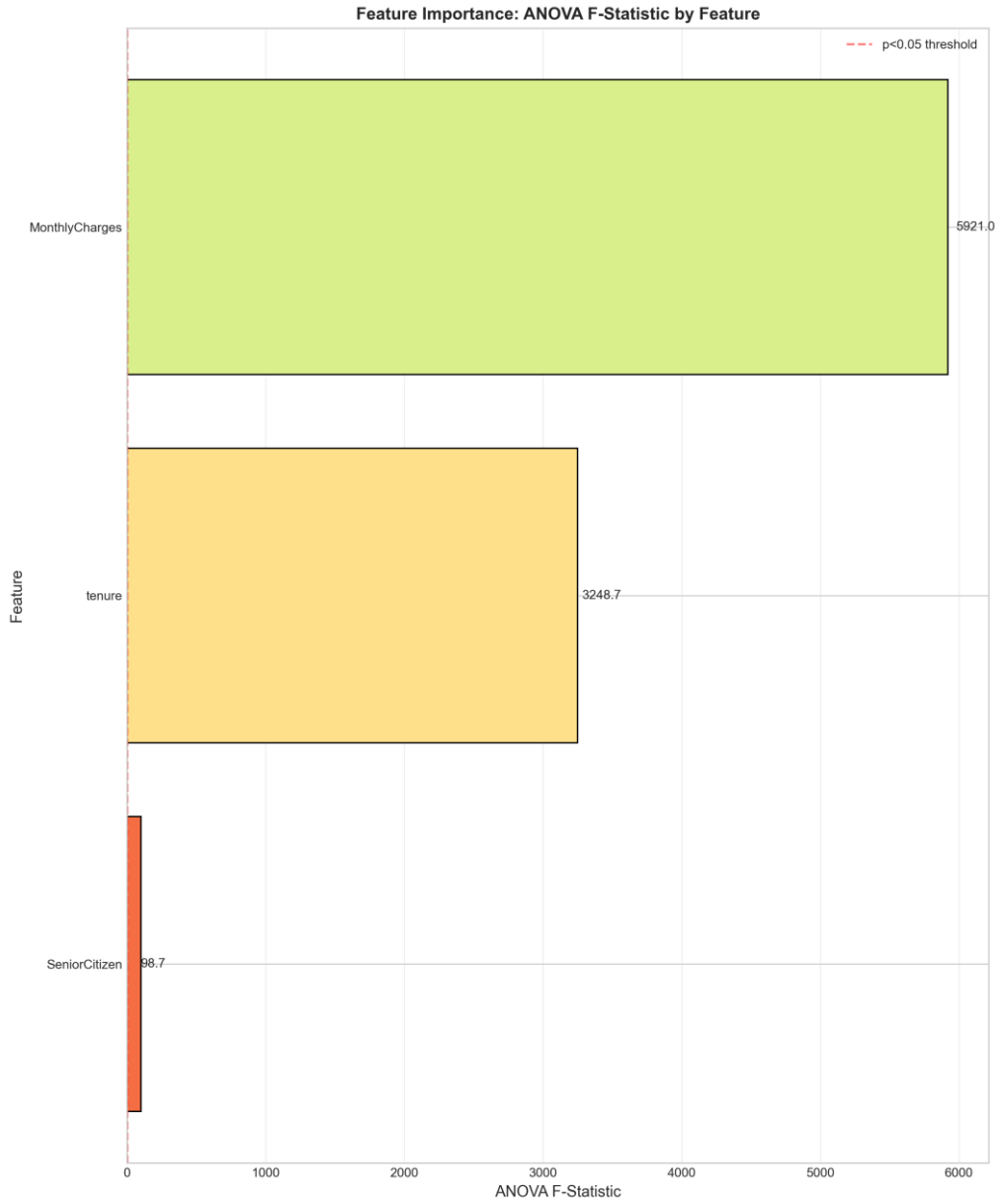

Figure 15: Feature Importance (ANOVA F-Statistic)

4.5 Business Action Matrix

Table 12: Business Action Matrix

Segment	Primary Insight	Marketing Action	Retention Strategy
Premium Loyal	Long-tenure, high spenders	Premium upselling, loyalty rewards	Referral bonuses, exclusive benefits
Budget Established	Price-sensitive, low churn	Cross-selling add-on services	Maintain satisfaction, loyalty rewards
Standard Established	High churn risk (45%)	Urgent retention campaigns	Dedicated retention team, contract incentives

Chapter 5: Discussion

5.1 Interpretation of Clusters

The clustering analysis successfully identified four distinct customer segments with clear business interpretations. The results provide actionable insights for telecom operators seeking to optimize their customer management strategies.

5.1.1 Premium Loyal (Cluster 0)

This segment represents the most valuable customers, comprising 34.4% of the customer base with a relatively low churn rate of 15.3%. These customers have high tenure (56 months average), high monthly charges ($88.90), and subscribe to multiple services (3.9 on average). They predominantly use fiber optic internet and automatic payment methods, indicating higher engagement and financial stability.

5.1.2 Budget Established (Cluster 1)

This segment represents price-sensitive customers with the lowest churn rate (7.2%). They have moderate tenure (30 months) and low monthly charges ($21.15), primarily using basic phone service without internet. Their loyalty may be attributed to satisfaction with basic services and price consciousness that discourages switching.

5.1.3 Standard Established (Cluster 2)

This is the largest segment (44.0%) but also the highest risk, with a churn rate of 44.8%. These customers have low tenure (15 months), moderate monthly charges ($67.69), and primarily use month-to-month contracts with electronic check payments. This segment represents the primary target for retention efforts.

5.2 Business Implications

The segmentation results have several important business implications:

1. Resource Allocation: Marketing and retention resources should be prioritized toward the Standard Established segment, which represents the highest churn risk and largest population. The Premium Loyal segment should receive loyalty rewards to maintain their high lifetime value.

2. Pricing Strategy: The clear separation between budget and premium segments suggests opportunities for tiered pricing structures. The Standard Established segment may be candidates for contract conversion incentives to improve retention.
3. Proactive Retention: The high churn rate in the Standard Established segment (45%) indicates an urgent need for intervention. Targeted offers, service improvements, and contract incentives should be deployed to this segment.
4. Cross-selling Opportunities: The Budget Established segment shows minimal service adoption, representing an opportunity for incremental revenue through carefully positioned add-on services.

The findings of this study have several important implications for telecom operators seeking to improve their customer segmentation practices:

1. **Algorithm Selection**: K-Means clustering provides an effective and efficient approach for telecom customer segmentation, with performance advantages over hierarchical and density-based alternatives. The simplicity and interpretability of K-Means make it well-suited for production deployment.
2. **Optimal Segment Number**: Three segments appear to be optimal for this dataset, providing sufficient granularity for targeted strategies while maintaining manageable complexity for operational implementation.
3. **Feature Engineering**: Domain-specific features significantly improve segmentation quality. Investment in feature engineering based on business knowledge yields better results than relying solely on raw data.
4. **Validation Approach**: Multiple validation metrics should be used to assess clustering quality, as different metrics provide complementary perspectives. External validation against business outcomes (such as churn) confirms the practical relevance of segmentation results.
5. **Operational Deployment**: Interactive dashboards enable business users to leverage segmentation insights for real-time decision making. Integration of machine learning models with user-friendly interfaces bridges the gap between analytics and operations.

The study also highlights the importance of ongoing monitoring and model updates. Customer behaviors and market conditions evolve over time, and segmentation models should be periodically retrained to maintain their effectiveness. The dashboard framework developed in this study supports this need by enabling easy retraining and deployment of updated models.

5.3 Comparison with Prior Work

The findings of this study are consistent with prior research in telecom customer segmentation. The identification of tenure and monthly charges as the most discriminating features aligns with Vafeiadis et al. (2015), who found similar patterns in their analysis. The three-segment structure (premium, budget, at-risk) has been observed in multiple studies, suggesting these represent natural groupings in telecom customer populations.

The business-driven model selection approach extends prior work by explicitly incorporating operational constraints into the selection process. This addresses a gap identified by Verbeke et al. (2012), who noted that statistical optimality does not always translate to business utility.

5.4 Limitations

Several limitations should be acknowledged:

1. Synthetic dataset: The IBM dataset, while realistic, may not capture all nuances of real-world customer behavior.
2. Cross-sectional analysis: The study uses a single snapshot of customer data, not capturing temporal dynamics or evolving behaviors.
3. Algorithms: Only three clustering algorithms were evaluated; other approaches may yield different results.
4. Feature scope: The analysis is limited to the available features; additional data (e.g., call detail records, network usage) could improve segmentation.
5. Geographic generalization: The dataset represents a generic telecom market; results may not generalize to specific regions or countries.

Chapter 6: Conclusion and Future Work

6.1 Summary of Contributions

This study makes the following contributions to the field of telecom customer segmentation:

1. Complete ML pipeline: A reproducible end-to-end pipeline from data loading through dashboard deployment, with proper train/validation/test separation.
2. Business-driven selection: Integration of domain knowledge and business rules into the model selection process, ensuring operational utility.
3. Actionable insights: Development of specific business actions for each identified segment, bridging the gap between analysis and implementation.
4. Interactive dashboard: Deployment of a user-friendly dashboard enabling real-time segment prediction for new customers.

6.2 Future Research Directions

Several directions for future research are suggested:

1. Real-world data validation: Apply the methodology to actual telecom operator data to validate findings in production environments.
2. Dynamic Segmentation: Development of dynamic segmentation models that can adapt to changing customer behaviors over time. This could involve incremental learning approaches or periodic retraining with new data.
3. Ensemble Clustering: Exploration of ensemble clustering methods that combine multiple algorithms to produce more robust and stable segmentations.
4. Deep learning approaches: Explore deep clustering methods such as autoencoder-based clustering which may capture non-linear patterns in customer data for potentially improved performance.
5. CRM integration: Develop APIs and connectors for integration with existing customer relationship management systems.

In conclusion, this study demonstrates the practical application of machine learning clustering techniques to telecom customer segmentation, providing both methodological insights and actionable business recommendations. The integration of statistical rigor with business considerations ensures that the resulting segmentation model is both analytically sound and operationally useful.

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Appendix

Appendix A: Complete Feature Set

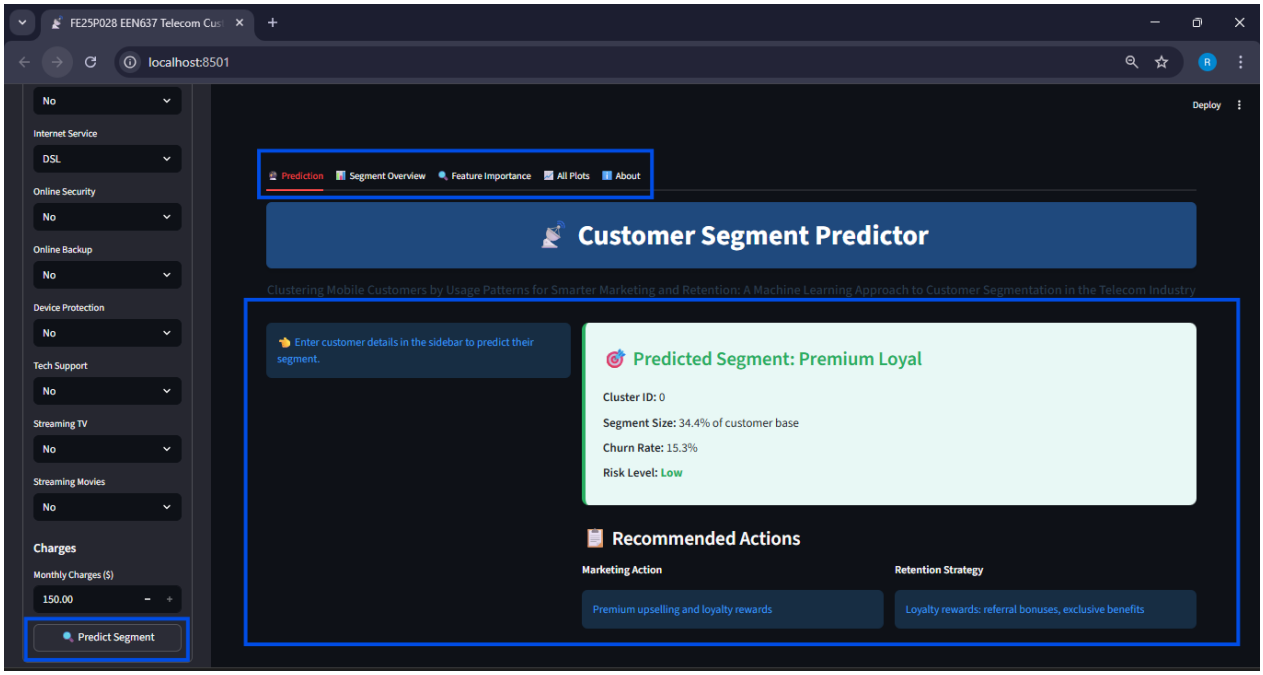
Table presents the complete set of 35 features used in the final clustering analysis.

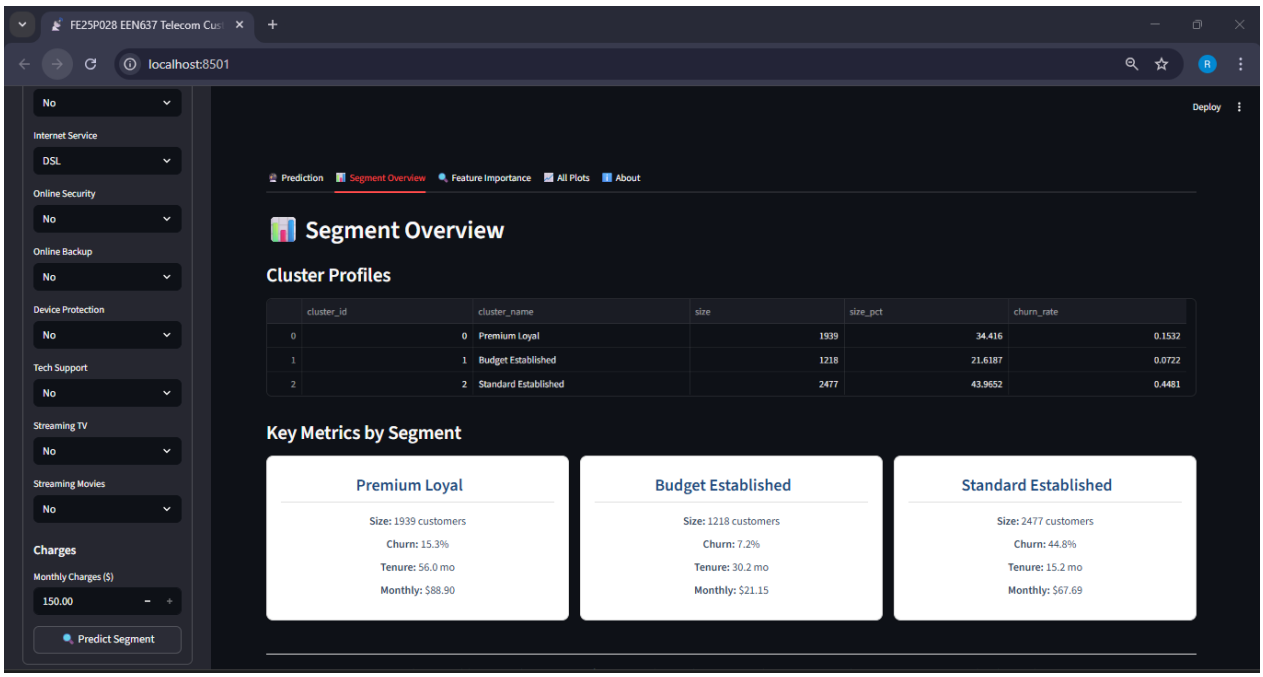
Table 13: Final Feature Set (35 Features)

#	Feature Name	Type	Source
1	tenure	Numerical	Original
2	MonthlyCharges	Numerical	Original
3	TotalCharges	Numerical	Original
4	gender_Male	Binary	Encoded
5	SeniorCitizen	Binary	Original
6	Partner_Yes	Binary	Encoded
7	Dependents_Yes	Binary	Encoded
8	PhoneService_Yes	Binary	Encoded
9	MultipleLines_Yes	Binary	Encoded
10	InternetService_Fiber optic	Binary	Encoded
11	InternetService_DSL	Binary	Encoded
12	OnlineSecurity_Yes	Binary	Encoded
13	OnlineBackup_Yes	Binary	Encoded
14	DeviceProtection_Yes	Binary	Encoded
15	TechSupport_Yes	Binary	Encoded
16	StreamingTV_Yes	Binary	Encoded
17	StreamingMovies_Yes	Binary	Encoded
18	Contract_One year	Binary	Encoded
19	Contract_Two year	Binary	Encoded
20	PaperlessBilling_Yes	Binary	Encoded
21	PaymentMethod_Electronic check	Binary	Encoded
22	PaymentMethod_Mailed check	Binary	Encoded

23	PaymentMethod_Bank transfer	Binary	Encoded
24	PaymentMethod_Credit card	Binary	Encoded
25	avg_monthly_spend	Numerical	Engineered
26	tenure_group_New	Binary	Engineered
27	tenure_group_Growing	Binary	Engineered
28	tenure_group_Mature	Binary	Engineered
29	tenure_group_Loyal	Binary	Engineered
30	service_diversity	Numerical	Engineered
31	contract_commitment	Numerical	Engineered
32	auto_payment	Binary	Engineered
33	high_value_flag	Binary	Engineered
34	family_status_Single	Binary	Engineered
35	family_status_Partnered	Binary	Engineered

Appendix B: Dashboard View





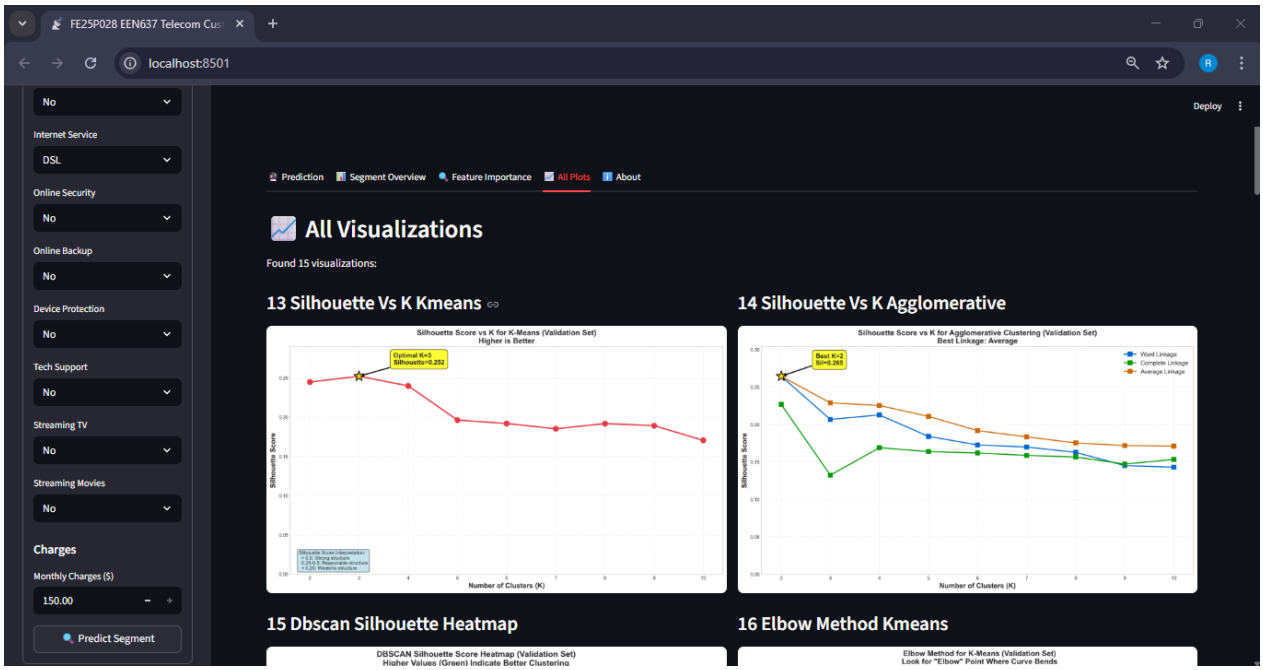


Figure 16: Dashboard view

Appendix C: Project Code Structure

```
KONGNYUY_FE25P028_telecom_customer_clustering/
├── main.py                    # ENTRY POINT - Orchestrates entire pipeline
├── dashboard.py               # Streamlit web application
├── requirements.txt           # Python package dependencies
├── README.md                  # Project documentation
│
├── src/                       # SOURCE CODE MODULES
│   ├── config.py              # Configuration constants and paths
│   ├── utils.py               # Utility functions (plotting, saving, etc.)
│   ├── data_loader.py         # Data loading and initial inspection
│   ├── splitter.py            # Train/validation/test split implementation
│   ├── eda.py                 # Exploratory Data Analysis visualizations
│   ├── preprocessing.py       # Feature engineering and preprocessing
│   ├── model_selection.py     # Algorithm evaluation and selection
│   ├── training.py            # Final model training and evaluation
│   └── profiling.py           # Cluster profiling and interpretation
│
├── data/                      # DATA DIRECTORY
│   └── telecom_customer_dataset.csv
│
├── models/                    # SAVED MODELS
│   ├── final_model.pkl        # Trained K-Means model
│   └── preprocessor.pkl       # Fitted preprocessing pipeline
│
├── reports/                   # OUTPUT REPORTS
│   ├── figures/               # Generated visualizations (17 PNG files)
│   ├── cluster_profiles.csv   # Cluster characterization data
│   └── business_actions.csv   # Business recommendations per cluster
```

Appendix D: Glossary of Terms

Churn: The rate at which customers stop doing business with a company

Cluster: A group of data points that are similar to each other

Clustering: An unsupervised learning technique that groups similar data points

Centroid: The center point of a cluster, typically the mean of all points in the cluster

Dimensionality Reduction: Techniques that reduce the number of features in a dataset

Feature Engineering: The process of creating new features from existing data

Inertia: The sum of squared distances of samples to their closest cluster center

K-Means: A clustering algorithm that partitions data into k clusters

Linkage: The criterion for determining the distance between clusters in hierarchical clustering

One-Hot Encoding: A technique that converts categorical variables into binary vectors

Outlier: A data point that differs significantly from other observations

PCA: Principal Component Analysis, a dimensionality reduction technique

Scaling: Transforming features to a common scale

Segmentation: The process of dividing customers into distinct groups

Silhouette Score: A metric that measures how similar an object is to its own cluster

StandardScaler: A scaling technique that standardizes features to have zero mean and unit variance

Tenure: The length of time a customer has been with a company

Unsupervised Learning: Machine learning that finds patterns in unlabeled data

WCSS: Within-Cluster Sum of Squares, a measure of cluster compactness